Value Chain GHG Footprint Analysis

Surf Corporation

Scope 1, 2 & 3 Greenhouse Gas Emissions Assessment

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1. Introduction

Surf Corporation is a global consumer products company headquartered in Easton, Delaware, with annual revenues of $10.8 billion USD. Originally founded in 1947 as the Surfactant Chemical Company, the organization provides personal and home cleaning products across over 100 countries, operating 18 manufacturing facilities with approximately 22,000 full-time employees and annual production of roughly 3.5 million metric tonnes.

Historically, Surf Corporation's sustainability efforts have focused primarily on improving climate performance related to its own buildings and physical operations through energy efficiency measures. While these internal sustainable operations strategies remain important, growing demands from internal stakeholders, external NGOs, and investor inquiries have escalated the profile of sustainability with the company's Board of Directors and Senior Management.

This report presents a comprehensive greenhouse gas (GHG) footprint analysis covering direct emissions (Scope 1), indirect energy emissions (Scope 2), and selected value chain emissions (Scope 3). The objective is to provide a prioritized evaluation of where the organization should focus its GHG reduction efforts beyond current facility operations, using industry-accepted practices aligned with the GHG Protocol Corporate Standard.

2. Methodology

The analysis follows the GHG Protocol Corporate Accounting and Reporting Standard framework, which categorizes emissions into three scopes:

- **Scope 1 (Direct Emissions):** GHG emissions from sources owned or controlled by the company, specifically on-site fuel combustion (natural gas and diesel) at manufacturing facilities.
- **Scope 2 (Indirect Energy Emissions):** GHG emissions from purchased electricity consumed at manufacturing facilities across five countries, using location-based emission factors.
- **Scope 3 (Value Chain Emissions):** Indirect emissions from four categories: Category 1 (Purchased Goods & Services), Category 3 (Fuel & Energy Related Activities), Category 6 (Business Travel), and Category 11 (Use of Sold Products).

Emission factors were sourced from the IEA, US EPA, and US EPA EEIO database. Electricity grid factors correspond to the most recent year available (2022). All results are expressed in metric tonnes of CO_2 equivalent (MT CO_2e).

3. Scope 1: Direct Emissions

Scope 1 emissions arise from the direct combustion of natural gas and diesel fuel across Surf Corporation's 18 manufacturing facilities. These fuels are used for process heating, equipment operation, and facility maintenance.

Source	Activity Data	Emission Factor	Emissions (MT CO_2e)
Natural Gas	12,010,199 m^3/yr	1.8868 $kgCO_2e/m^3$	22,660.84
Diesel Fuel	1,392,598 litres/yr	2.68566 $kgCO_2e$/litre	3,740.04
Total Scope 1			**26,400.89**

Natural gas accounts for approximately 86% of direct emissions, with diesel making up the remaining 14%. Total Scope 1 emissions amount to 26,400.89 MT CO_2e, representing the smallest portion of Surf Corporation's overall carbon footprint.

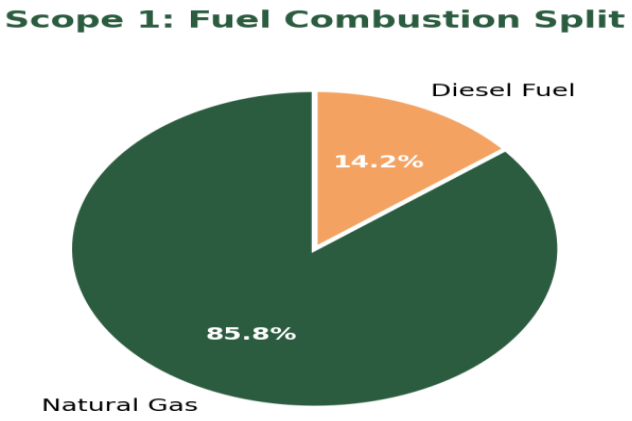

Figure 1: Scope 1 emissions split by fuel type

4. Scope 2: Indirect Energy Emissions

Scope 2 emissions result from purchased electricity consumed at manufacturing facilities across five countries. Location-based emission factors from the IEA (international) and US EPA eGRID (United States) databases for 2022 were applied.

Country / Grid	Usage (MWh/yr)	EF ($kgCO_2e$/MWh)	Emissions (MT CO_2e)
China	150,500	735.45	110,685.28
Mexico	215,000	436.64	93,877.76
USA (Texas / ERCT)	50,500	516.50	26,083.44
France	179,500	75.93	13,629.15
South Africa	45,500	889.48	40,471.13
Total Scope 2	**641,000**		**284,746.75**

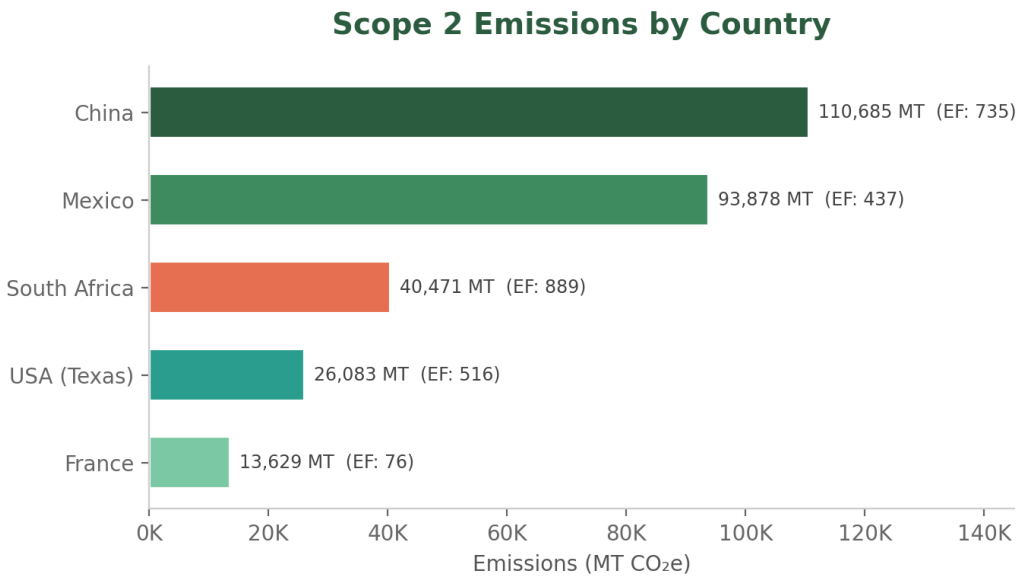

Figure 2: Scope 2 emissions by country with grid emission factors

China is the largest contributor to Scope 2 emissions (39%) despite not having the highest electricity consumption. This is because China's coal-heavy grid has a high emission factor. Mexico contributes 33% due to its combination of high usage and a moderately carbon-intensive grid. France, despite consuming the second-highest volume of electricity (179,500 MWh), contributes only 5% of Scope 2 emissions thanks to its predominantly nuclear power grid, which results in a very low emission factor of 75.93 $kgCO_2e/MWh$.

5. Scope 3: Value Chain Emissions

5.1 Category 1: Purchased Goods & Services

This category captures upstream emissions embedded in the goods and services Surf Corporation procures. Spend-based emission factors from the US EPA Environmentally Extended Input-Output (EEIO) model were applied to three major procurement categories.

Category	Spend (USD)	EF ($kgCO_2e$/USD)	Emissions (MT CO_2e)
Chemical Raw Materials	$1,200,000,000	1.164	1,396,800
Machinery	$100,000,000	0.117	11,700
Real Estate	$20,000,000	0.086	1,720
Total Category 1			**1,410,220**

Chemical raw materials (petrochemicals) overwhelmingly dominate this category at 99% of Category 1 emissions. This is driven by both the massive spend volume ($1.2B) and the high emission intensity of petrochemical manufacturing (1.164 $kgCO_2e$ per dollar). This finding directly reflects Surf Corporation's core business as a chemical-based consumer products company.

5.2 Category 3: Fuel & Energy Related Activities

This category captures indirect emissions associated with the extraction, production, and transportation of fuels and energy purchased by the organization but not already accounted for in Scope 1 or Scope 2. Based on the organizational profile, these emissions are estimated at approximately 10% of total Scope 1 and Scope 2 emissions combined.

Basis	Emissions (MT CO_2e)	Factor
Scope 1 Total	26,400.89	
Scope 2 Total	284,746.75	
Combined Scope 1 + 2	311,147.64	10%
Total Category 3	**31,114.76**	

5.3 Category 6: Business Travel

Business travel emissions encompass air travel and car travel conducted by Surf Corporation employees for business purposes.

Mode	Distance (miles)	EF (kgCO_2e/mile)	Emissions (MT CO_2e)
Air Travel	45,000,000	0.1639	7,377.60
Car Travel	2,830,000	0.3080	871.60
Total Category 6			**8,249.20**

Business travel represents the smallest Scope 3 category at 8,249.20 MT CO_2e. Air travel accounts for approximately 89% of travel emissions, while car travel contributes the remaining 11%.

5.4 Category 11: Use of Sold Products

This category estimates the downstream emissions generated when consumers use Surf Corporation's products. For cleaning and personal care products, emissions primarily arise from the energy required to heat water during product use. Emission factors are expressed as MT CO_2e per MT of product and reflect lifecycle assessment data for each product type.

Product	Volume (MT/yr)	EF (MT CO_2e/MT)	Emissions (MT CO_2e)
Soap & Shampoo	1,750,000	0.353	617,750
Laundry Detergent	1,050,000	0.2829	297,045
Dish Soap	650,000	0.504	327,600
Floor Cleaners	250,000	2.828	707,000
Total Category 11	**3,700,000**		**1,949,395**

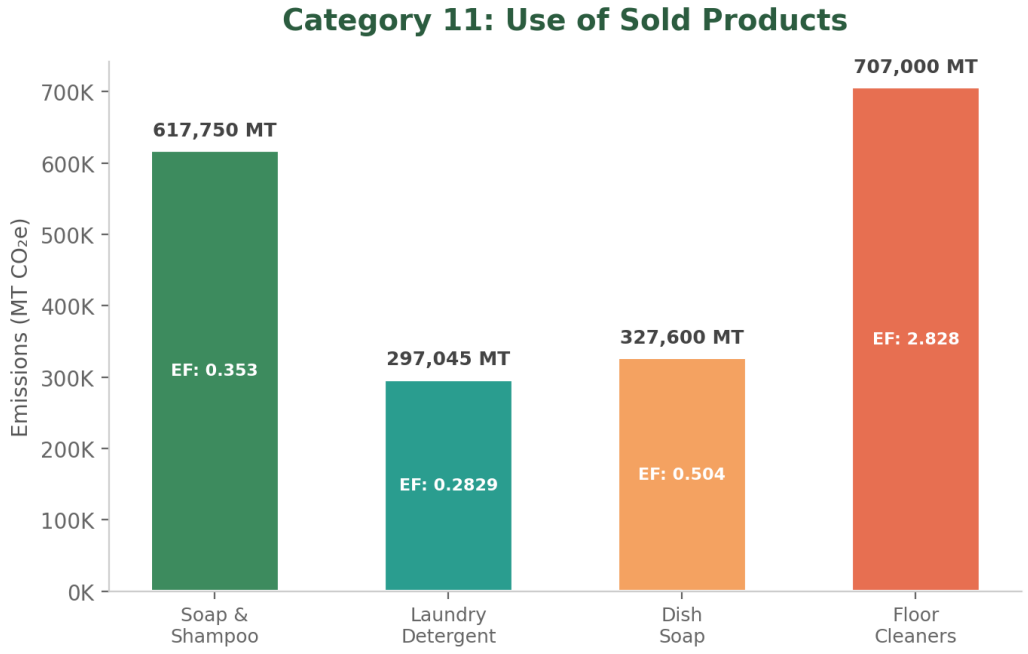

Figure 3: Category 11 emissions by product type with emission factors

Category 11 is the single largest emission source across all scopes at 1,949,395 MT CO_2e. Floor cleaners, despite having the lowest production volume (250,000 MT), contribute the most emissions (36%) due to their dramatically higher emission factor of 2.828 MT CO_2e/MT, roughly 8 times higher than soap and shampoo. This likely reflects the greater energy intensity of floor cleaning activities (hot water, machine use).

6. Consolidated GHG Footprint Summary

Emissions Category	Emissions (MT CO_2e)	% of Total
Scope 1: Direct Emissions	26,401	0.7%
Scope 2: Purchased Electricity	284,747	7.7%
Scope 3, Cat 1: Purchased Goods/Services	1,410,220	38.0%
Scope 3, Cat 3: Fuel/Energy Activities	31,115	0.8%
Scope 3, Cat 6: Business Travel	8,249	0.2%
Scope 3, Cat 11: Use of Sold Products	1,949,395	52.5%
Total Scope 3	**3,398,979**	**91.6%**
TOTAL GHG FOOTPRINT	**3,710,127**	**100%**

7. Key Findings and Recommendations

The GHG footprint analysis reveals several critical insights for Surf Corporation's sustainability strategy:

1. **Scope 3 dominates the footprint.**

Value chain emissions account for 91.6% of total GHG emissions. Surf Corporation's current focus on facility energy efficiency, while valuable, addresses less than 9% of the overall footprint. Any meaningful decarbonization strategy must extend beyond Scope 1 and 2 boundaries.

2. **Use of Sold Products is the single largest category.**

At 52.5% of total emissions, Category 11 represents the most significant impact area. Consumers using Surf Corporation's products generate nearly 2 million MT CO_2e annually, primarily through water heating during product use. Floor cleaners have an outsized impact relative to their production volume due to high use-phase energy requirements.

3. **Purchased Goods & Services is the second largest driver.**

At 38.0%, petrochemical raw material procurement contributes over 1.4 million MT CO_2e. This reflects the carbon-intensive nature of the petrochemical supply chain and presents a significant engagement opportunity with upstream suppliers.

4. **Grid decarbonization can reduce Scope 2.**

The disparity in electricity emission factors across operating countries (France at 75.93 vs. South Africa at 889.48 $kgCO_2e/MWh$) highlights opportunities to reduce Scope 2 emissions through renewable energy procurement, particularly in China, Mexico, and South Africa where grid factors are highest.

Recommended Priority Areas

- **Product reformulation and concentrated product formats** to reduce use-phase emissions, particularly for floor cleaners and dish soap where emission factors are highest.
- **Supplier engagement on petrochemical sourcing** through science-based targets, bio-based alternatives, and supply chain emissions reporting requirements.
- **Renewable energy procurement** for high-impact operating countries, especially China and South Africa, through power purchase agreements (PPAs) or renewable energy certificates (RECs).
- **Cold-water product innovation** to reduce consumer energy use during the product use phase, which would directly address the largest emissions category.

8. Conclusion

This analysis demonstrates that Surf Corporation's true carbon footprint extends far beyond its facility walls. At 3.71 million MT CO_2e, the company's total emissions are dominated by value chain activities, with the use phase of sold products and petrochemical procurement together accounting for over 90% of total emissions. This pattern is consistent with consumer products companies globally, where Scope 3 typically represents the vast majority of the GHG footprint.

The findings underscore the need for Surf Corporation to evolve its sustainability strategy from a facility-focused approach to a comprehensive value chain decarbonization program. By targeting product innovation, supplier engagement, and renewable energy procurement, the company can align its sustainability efforts with where the actual emissions are and deliver on its stated commitment to Responsible Growth.

GHG Calculation Excel Sheet - https://github.com/devangnajadeja/Value-Chain-GHG-Footprint-Analysis/blob/main/GHG-Calc.xlsx